# Chapter 36

Diffraction

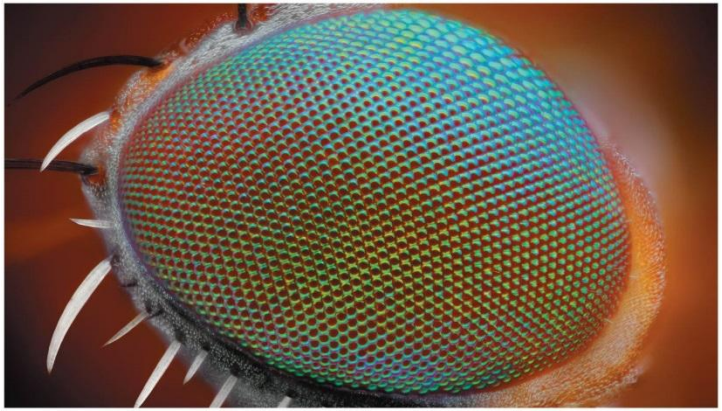
# Learning Outcomes

**In this chapter, you'll learn…**

- how to calculate the intensity at various points in a single-slit diffraction pattern.
- what happens when coherent light shines on an array of narrow, closely spaced slits.
- how x-ray diffraction reveals the arrangement of atoms in a crystal.
- how diffraction sets limits on the smallest details that can be seen with an optical system.
- how holograms work.

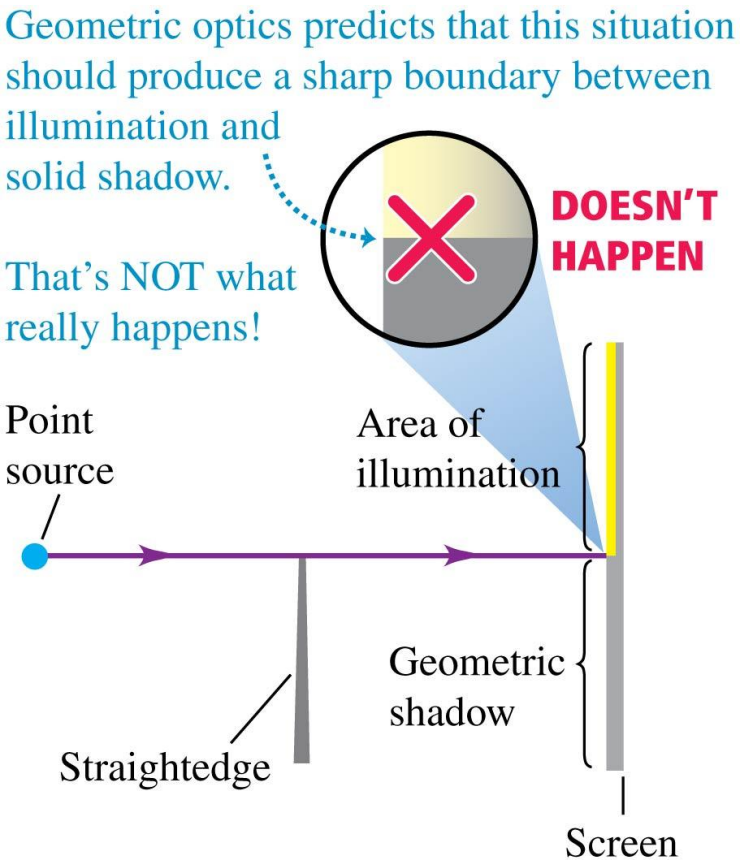
# Introduction

- Flies have compound eyes with thousands of miniature lenses, each only about 20 μm in diameter.
- Due to the wave-nature of light, the ability of a lens to resolve fine details improves as the lens diameter *D* increases.
- Each miniature lens in a fly's eye has very poor resolution, compared to those produced by a human eye, because the lens is so small.
- We'll continue our exploration of the wave nature of light with **diffraction**.



# Diffraction

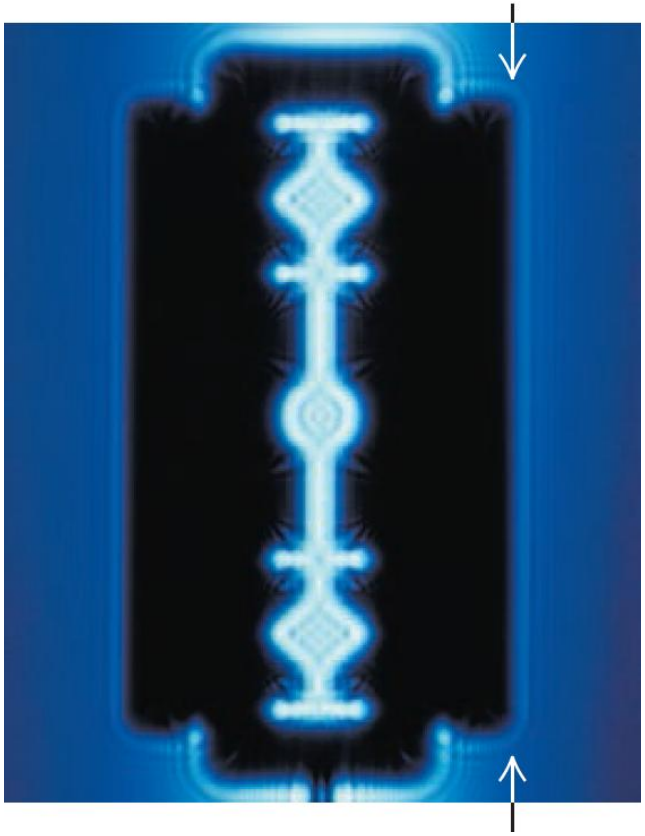
- According to geometric optics, when an opaque object is placed between a point light source and a screen, the shadow of the object forms a perfectly sharp line.
- However, the wave nature of light causes interference patterns, which blur the edge of the shadow.
- This is one effect of **diffraction**.



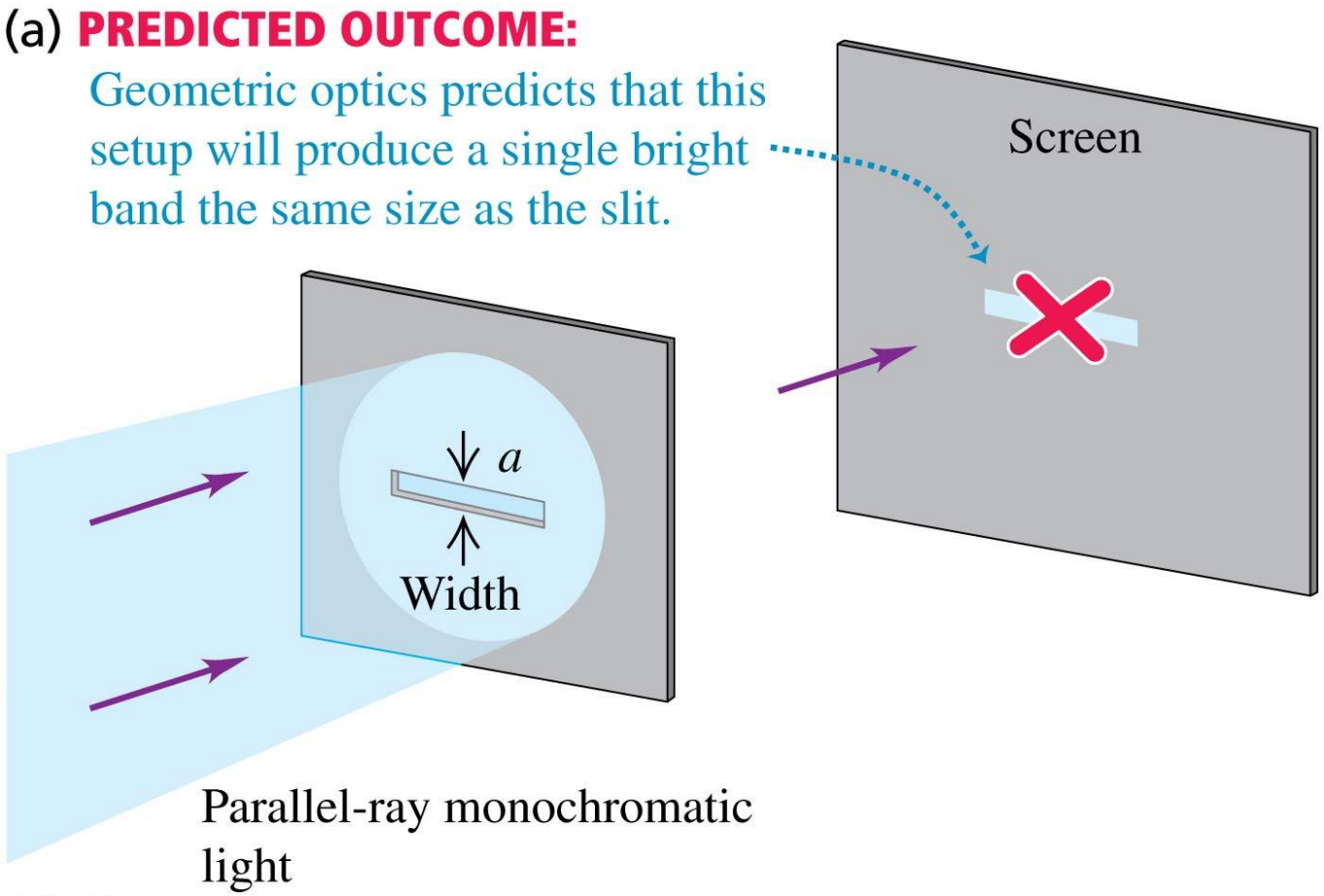
# Diffraction and Huygen's Principle

- This photograph was made by placing a razor blade halfway between a pinhole, illuminated by monochromatic light, and a photographic film.
- The film recorded the shadow cast by the blade.
- Note the fringe pattern around the blade outline, which is caused by diffraction.

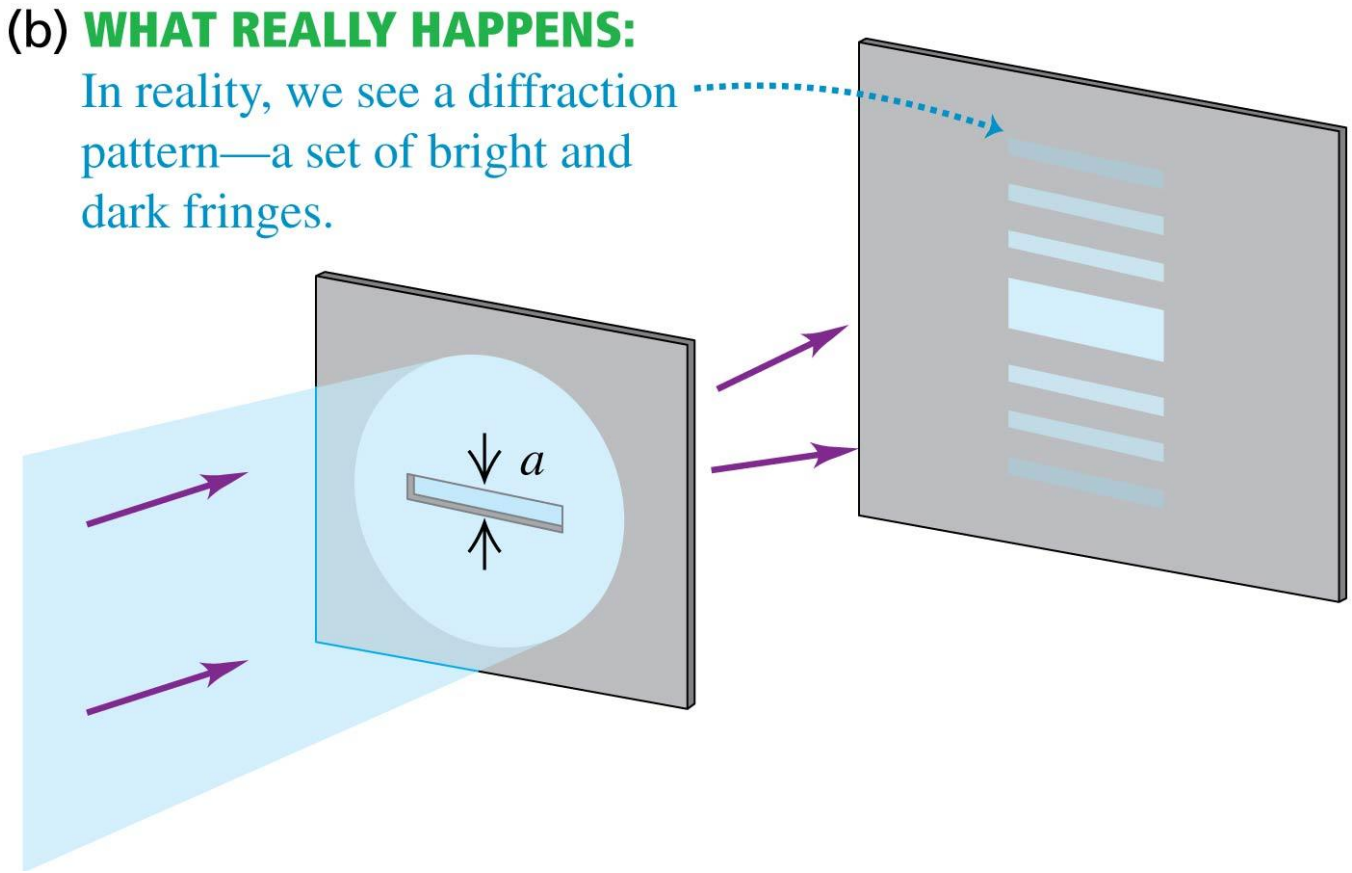
(a)



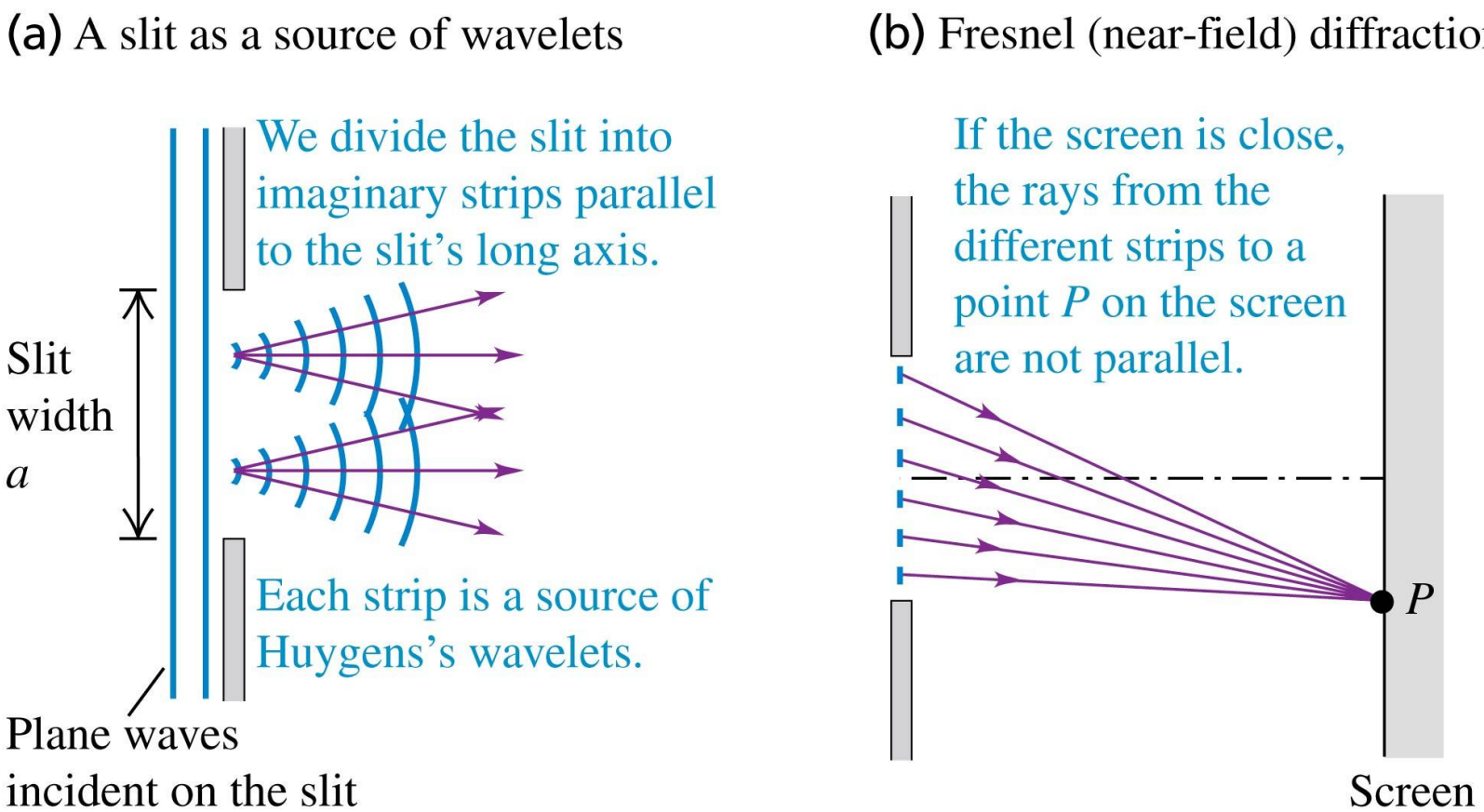
# Diffraction from a Single Slit (1 of 2)



# Diffraction from a Single Slit (2 of 2)

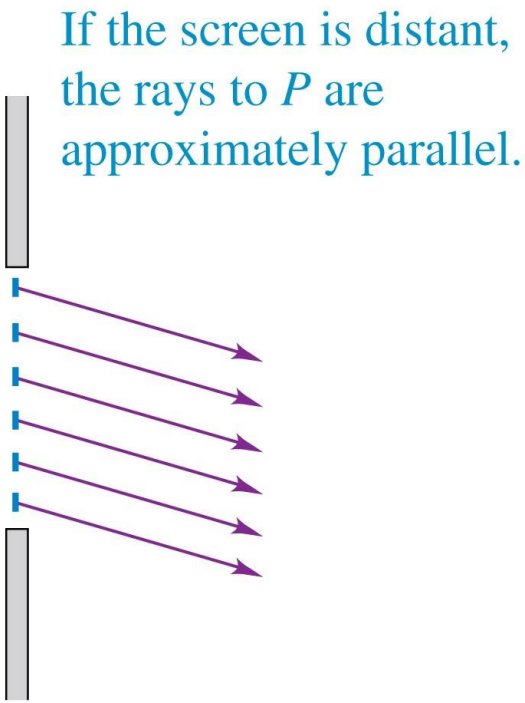


# Fresnel Diffraction by a Single Slit

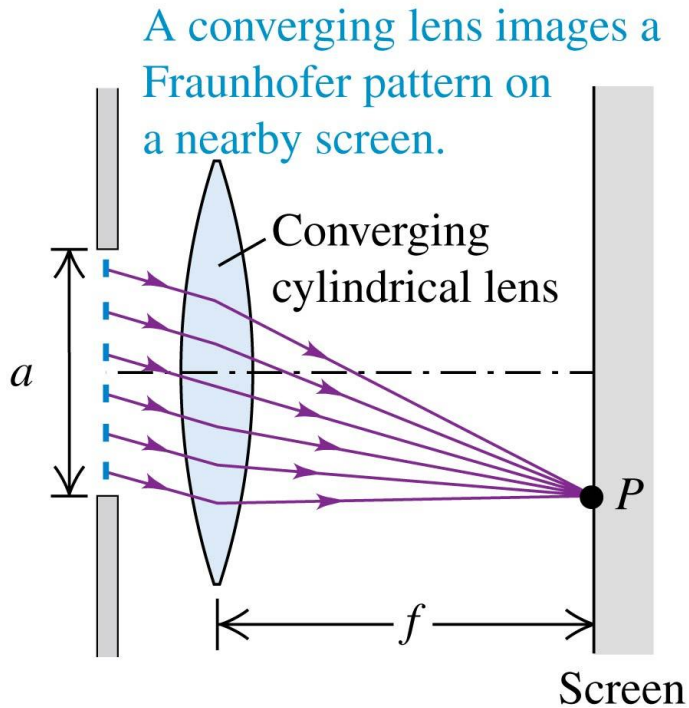


# Fraunhofer Diffraction by a Single Slit

(c) Fraunhofer (far-field) diffraction




(d) Imaging Fraunhofer diffraction



# Locating the Dark Fringes

- Shown is the Fraunhofer diffraction pattern from a single horizontal slit.
- It is characterized by a central bright fringe centered at $\theta = 0$, surrounded by a series of dark fringes.
- The central bright fringe is twice as wide as the other bright fringes.

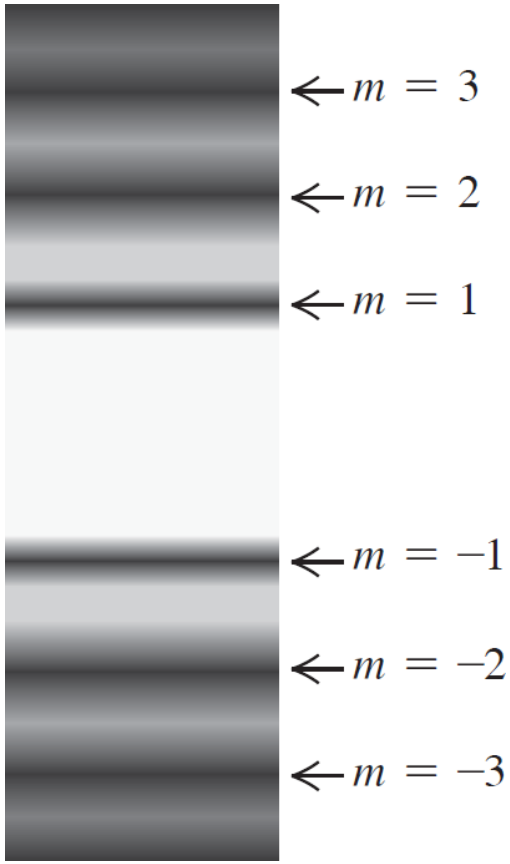
**Dark fringes, single-slit diffraction:**

Angle of line from center of slit to $m$th dark fringe on screen

$$\sin\theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \ldots)$$

Slit width

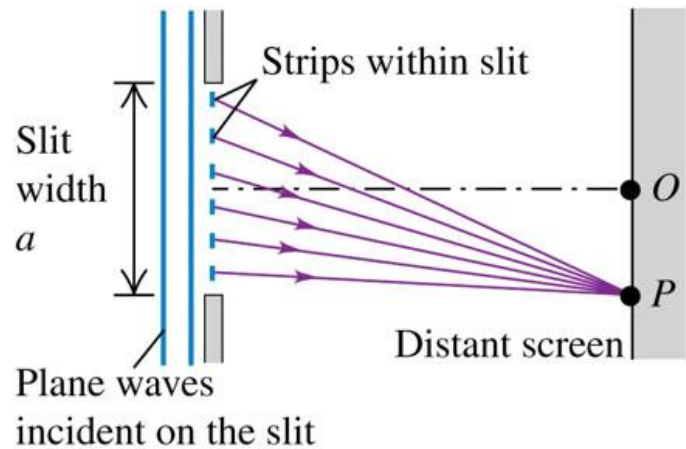
Wavelength



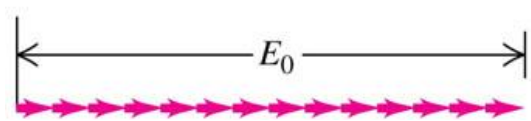

- Video Tutor Solution: Example 36.1

# Intensity in the Single-Slit Pattern (1 of 2)

- We can derive an expression for the intensity distribution for the single-slit diffraction pattern by using phasor-addition.
- We imagine a plane wave front at the slit subdivided into a large number of strips.
- At the point *O*, the phasors are all **in phase**.



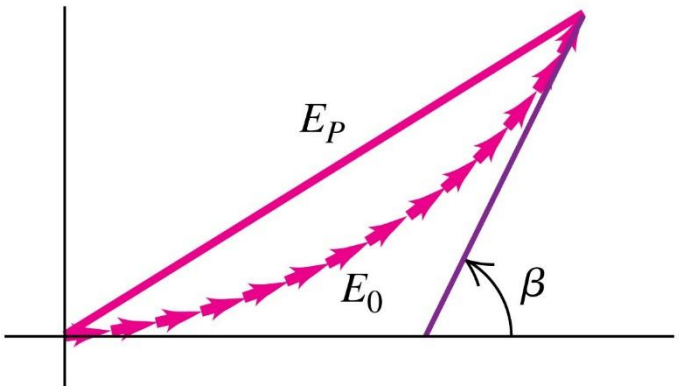

At the center of the diffraction pattern (point $O$), the phasors from all strips within the slit are in phase.



# Intensity in the Single-Slit Pattern (2 of 2)

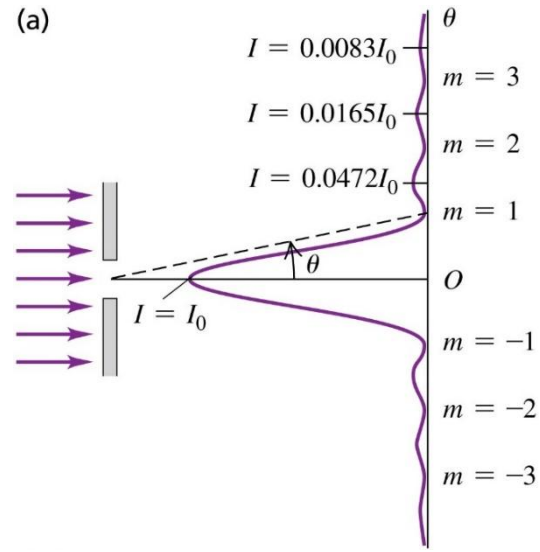
- Now consider wavelets arriving from different strips at point *P*.
- Because of the differences in path length, there are now phase differences between wavelets coming from adjacent strips.
- The vector sum of the phasors is now part of the perimeter of a many-sided polygon.

(c) Phasor diagram at a point slightly off the center of the pattern; $\beta$ = total phase difference between the first and last phasors.



# Intensity Maxima in a Single-Slit Pattern

- Shown is the intensity versus angle in a single-slit diffraction pattern.
- Most of the wave power goes into the central intensity peak (between the $m = 1$ and $m = -1$ intensity minima).




Angle of line from center of slit to position on screen

**Intensity in single-slit diffraction**

$$I = I_0 \left\{ \frac{\sin[\pi a(\sin\theta)/\lambda]}{\pi a(\sin\theta)/\lambda} \right\}^2$$

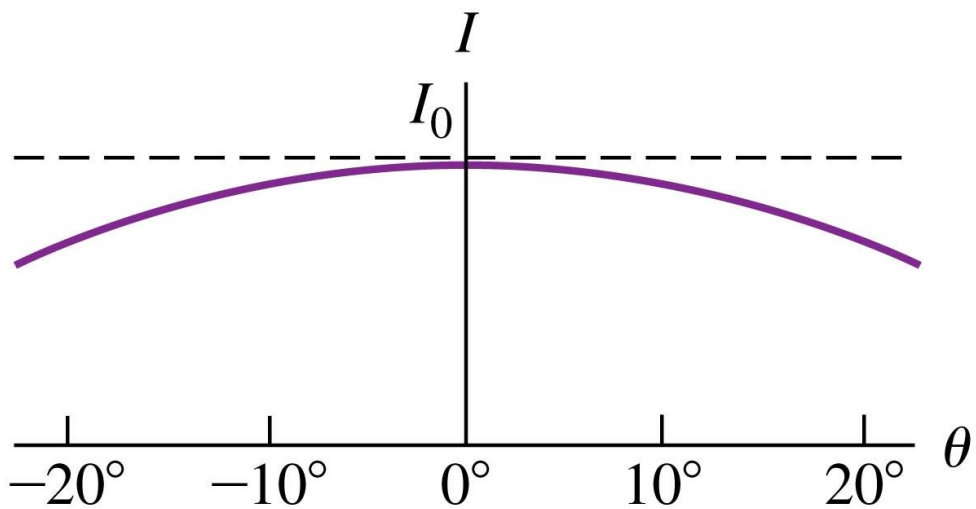
Intensity at $\theta = 0$ Slit width Wavelength

- Video Tutor Solution: Example 36.2

# Width of the Single-Slit Pattern (1 of 2)

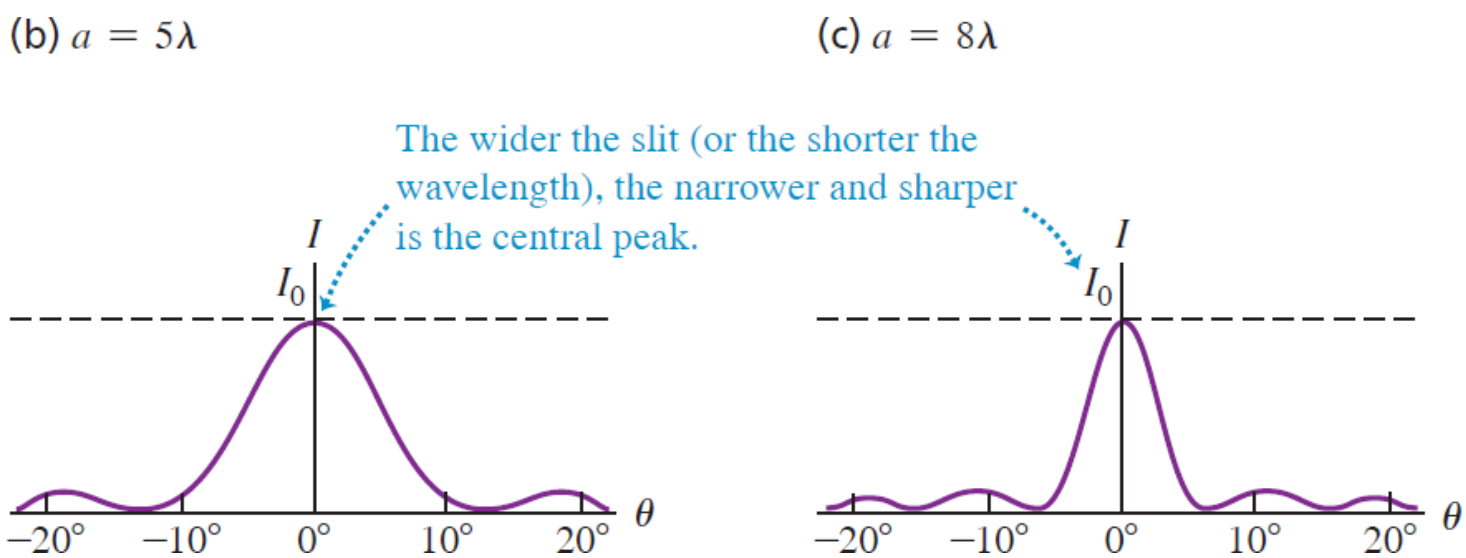
- The single-slit diffraction pattern depends on the ratio of the slit width *a* to the wavelength $\lambda$.
- Below is the pattern when $a = \lambda$.

If the slit width is equal to or narrower than the wavelength, only one broad maximum forms.



# Width of the Single-Slit Pattern (2 of 2)

- The single-slit diffraction pattern depends on the ratio of the slit width *a* to the wavelength $\lambda$.
- Below are the patterns when $a = 5\lambda$ (left) and $a = 8\lambda$ (right).



## Question 6

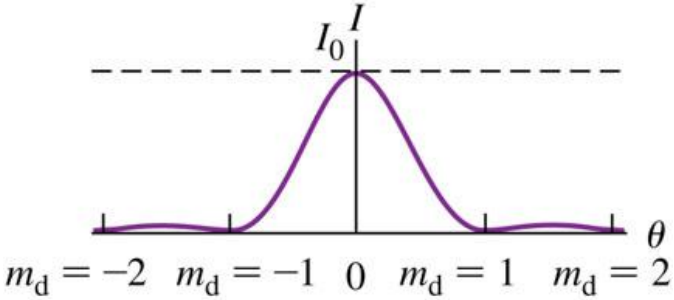
10 pts

A single slit experiment involves monochromatic, coherent light incident on an aperture of width 0.18 mm. If the screen is 1.8 m away, and the distance from the central maximum to the $m$ = 5 minimum is 8 mm, determine the wavelength of the light in nanometers.

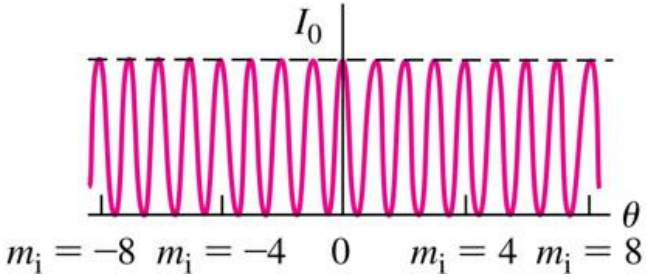
# Two Slits of Finite Width (1 of 2)

- Figure (a) shows the intensity in a single-slit diffraction pattern with slit width *a*.
- The diffraction minima are labeled by the integer $m_d = \pm 1, \pm 2, \ldots$ ("d" for "diffraction").
- Figure (b) shows the pattern formed by two very narrow slits with distance *d* between slits, where *d* is four times as great as the single-slit width *a*.
- "i" is for "interference."

(a) Single-slit diffraction pattern for a slit width *a*



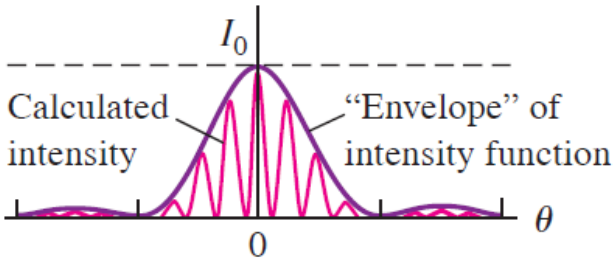

(b) Two-slit interference pattern for narrow slits whose separation *d* is four times the width of the slit in (a)



# Two Slits of Finite Width (2 of 2)

- Figure (c) shows the pattern from two slits with width *a*, separated by a distance (between centers) $d = 4a$.
- The two-slit peaks are in the same positions as before, but their intensities are modulated by the single-slit pattern, which acts as an "envelope" for the intensity function.
- Figure (d) shows the pattern, which is both from diffraction *and* interference.

(c) Calculated intensity pattern for two slits of width $a$ and separation $d = 4a$, including both interference and diffraction effects

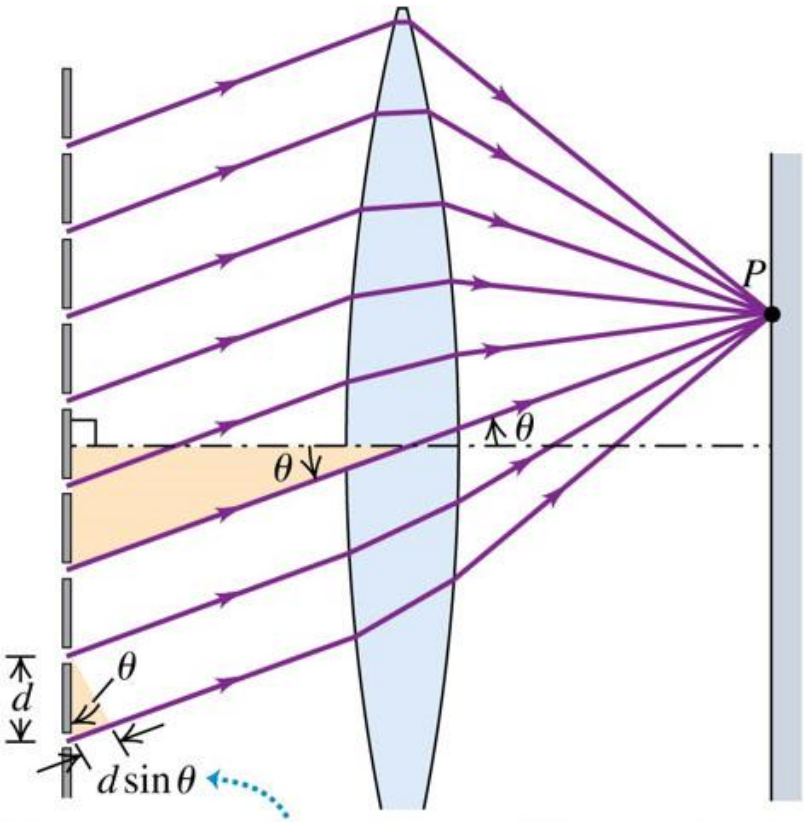



(d) Photograph of the pattern calculated in (c)

For $d = 4a$, every fourth interference maximum at the sides ($m_i = \pm 4, \pm 8, \ldots$) is missing.

# Several Slits

- Shown is an array of eight narrow slits, with distance *d* between adjacent slits.
- Constructive interference occurs for rays at angle $\theta$ to the normal that arrive at point *P* with a path difference between adjacent slits equal to an integer number of wavelengths.

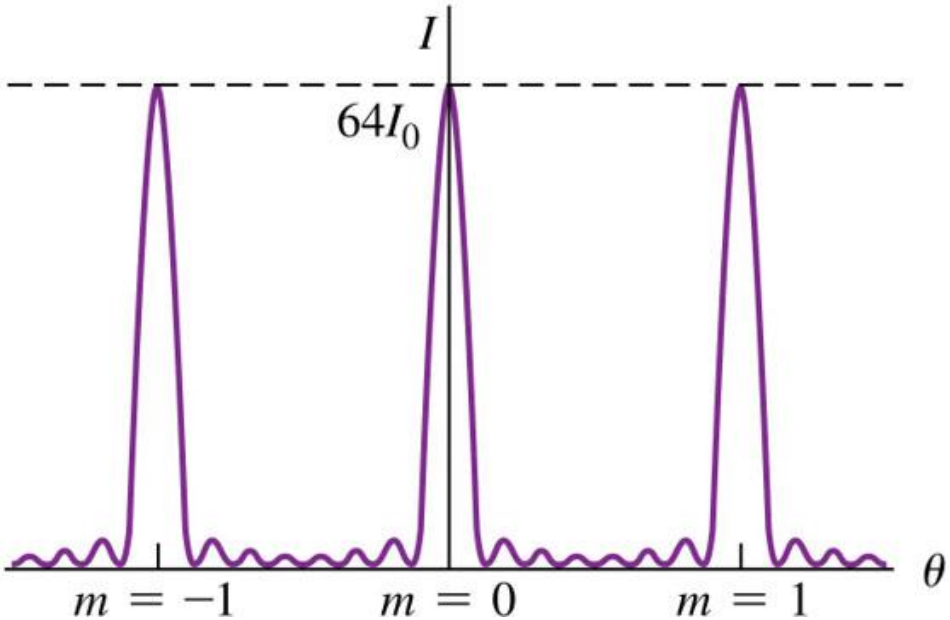



Maxima occur where the path difference for adjacent slits is a whole number of wavelengths: $d \sin\theta = m\lambda$.

# Interference Pattern of Several Slits (1 of 2)

- Shown is the result of a detailed calculation of the eight-slit pattern.
- The large maxima, called principal maxima, are in the same positions as for a two-slit pattern, but are much narrower.

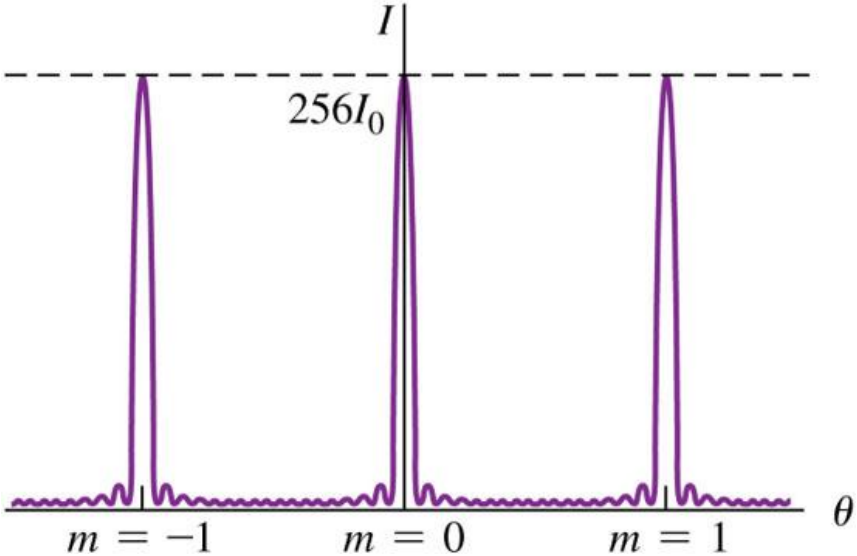
$N = 8$: eight slits produce taller, narrower maxima in the same locations, separated by seven minima.



# Interference Pattern of Several Slits (2 of 2)

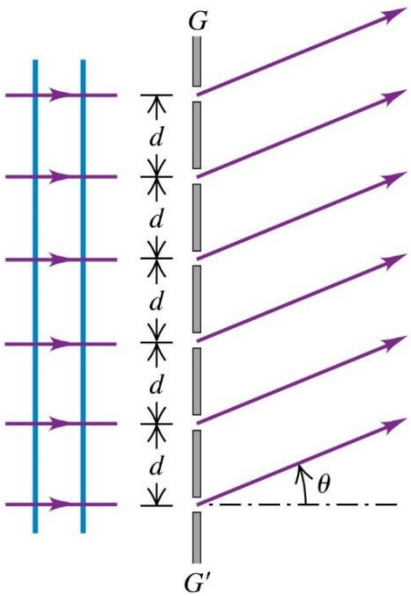
- Shown is the result for 16 slits.
- The height of each principal maximum is proportional to $N^2$, so from energy conservation, the width of each principal maximum must be proportional to $\frac{1}{N}$.

$N = 16$: with 16 slits, the maxima are even taller and narrower, with more intervening minima.



# The Diffraction Grating

- An array of a large number of parallel slits is called a **diffraction grating**.
- In the figure, $GG'$ is a cross section of a transmission grating.
- The slits are perpendicular to the plane of the page.
- The diagram shows only six slits; an actual grating may contain several thousand.




**Intensity maxima, multiple slits:**

$$d \sin\theta = m\lambda \qquad (m = 0, \pm 1, \pm 2, \ldots)$$

Distance between slits

Wavelength

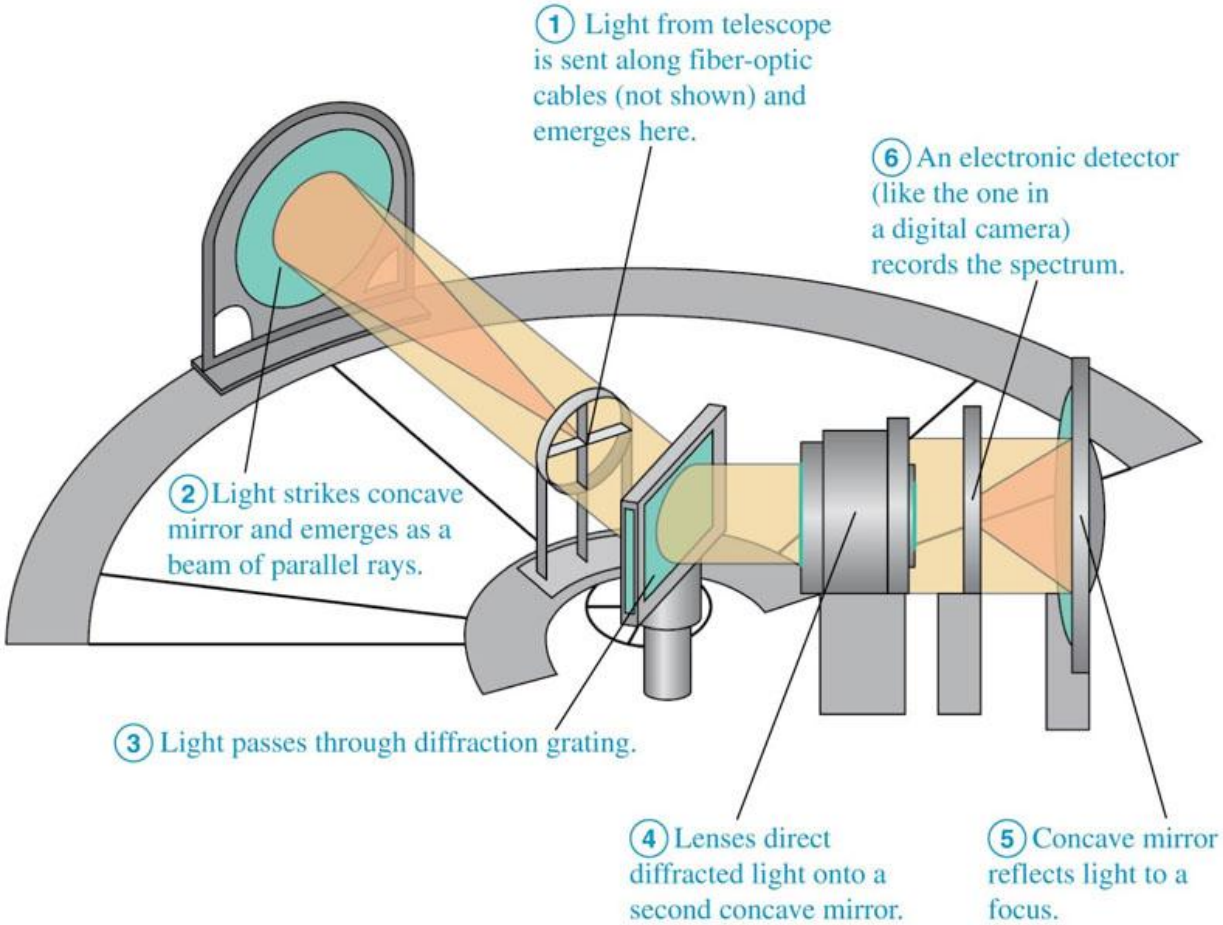
Angle of line from center of slit array to $m$th bright region on screen

- Video Tutor Solution: Example 36.4

# The Reflection Grating

- The rainbow-colored reflections from the surface of a DVD are a reflection-grating effect.
- The “grooves” are tiny pits 0.12 mm deep in the surface of the disc, with a uniform radial spacing of 0.74 mm = 740 nm.
- Information is coded on the DVD by varying the length of the pits.
- The reflection-grating aspect of the disc is merely an aesthetic side benefit.

# Diagram of a Grating Spectrograph



# Resolution of a Grating Spectrograph

- In spectroscopy it is often important to distinguish slightly differing wavelengths.
- The minimum wavelength difference $\Delta\lambda$ that can be distinguished by a spectrograph is described by the **chromatic resolving power** *R*.
- For a grating spectrograph with a total of *N* slits, used in the $m^{\text{th}}$ order, the chromatic resolving power is:

$$R = \frac{\lambda}{\Delta\lambda} = Nm$$

## Question 7

10 pts

The light emitted from an excited atom has two principal wavelengths: 589.22 nm and 588.5 nm. If you wish to resolve these two emission lines in first order using a diffraction grating that is 1.902 cm in length, what minimum number of slits per centimeter must the grating have?

## Question 8

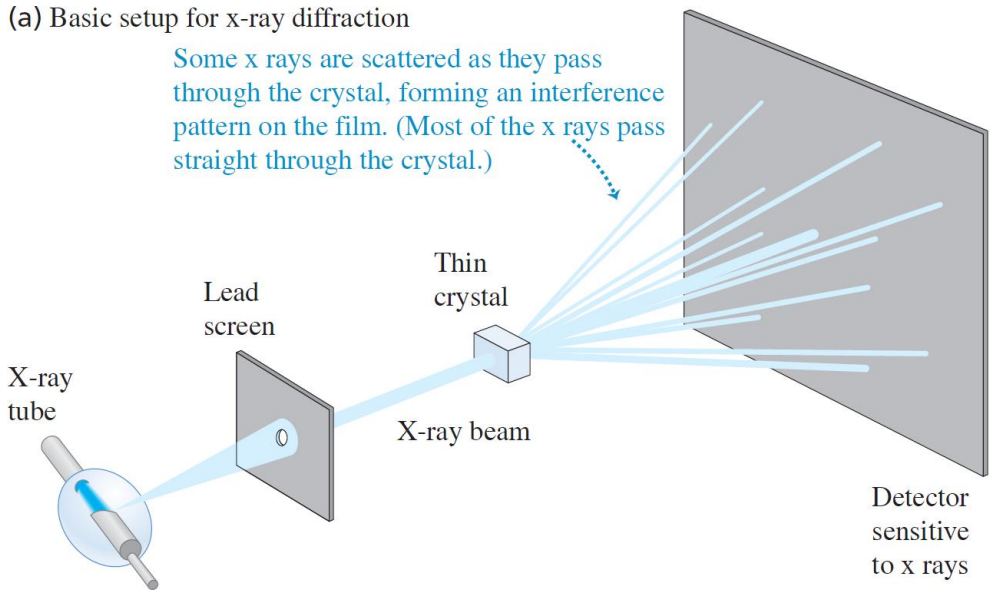
10 pts

A diffraction grating has 347 lines per centimeter. If monochromatic light of wavelength 553 nm is incident on the grating and the screen is 2.3 m away, determine the distance in cm from the center of the central maximum to the $m = 2$ maximum.

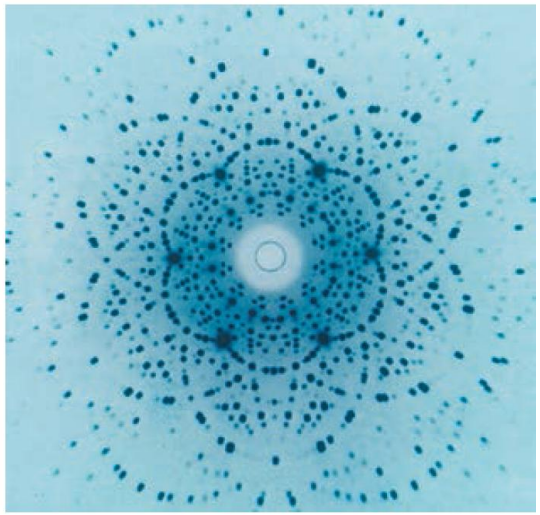
# X-Ray Diffraction

- When x rays pass through a crystal, the crystal behaves like a diffraction grating, causing **x-ray diffraction**.

(a) Basic setup for x-ray diffraction



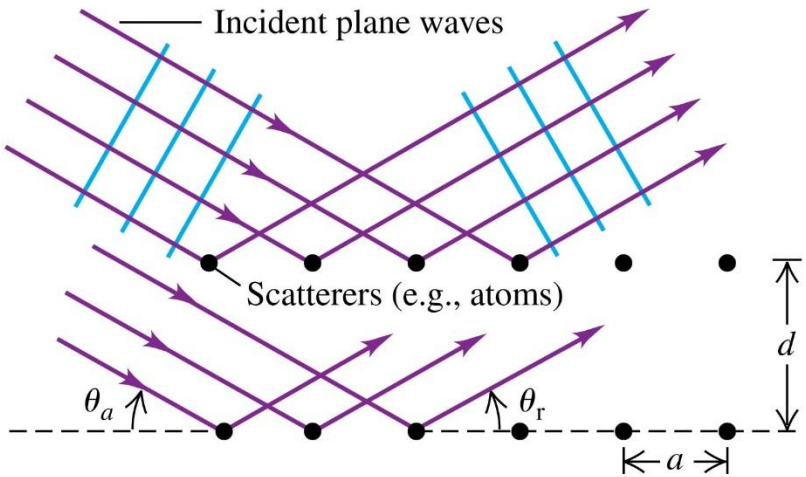

(b) Laue diffraction pattern for a thin section of beryl crystal



- Video Tutor Solution: Example 36.5

# A Simple Model of X-Ray Diffraction

- To better understand x-ray diffraction, we consider a two-dimensional scattering situation.
- The path length from source to observer is the same for all the scatterers in a single row if $\theta_a = \theta_r = \theta$.




**Bragg condition for constructive interference from an array:**

Distance between adjacent rows in array

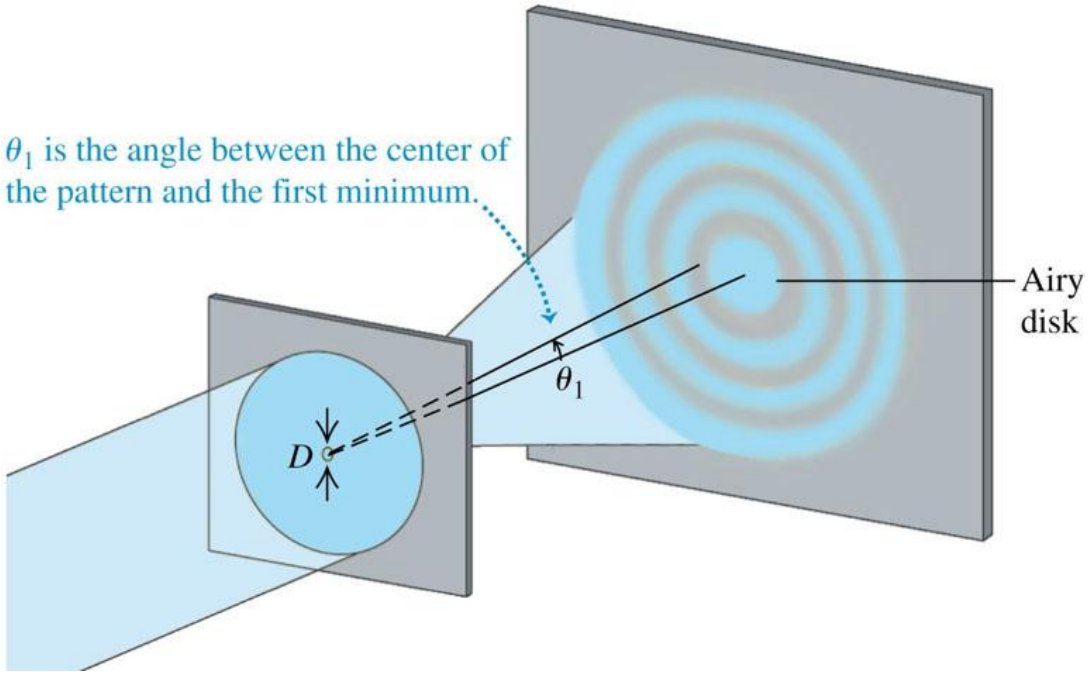
Wavelength

$$2d\sin\theta = m\lambda \qquad (m = 1, 2, 3, \ldots)$$

Angle of line from surface of array to $m$th bright region on screen

# Circular Apertures

- The diffraction pattern formed by a circular aperture consists of a central bright spot surrounded by a series of bright and dark rings.



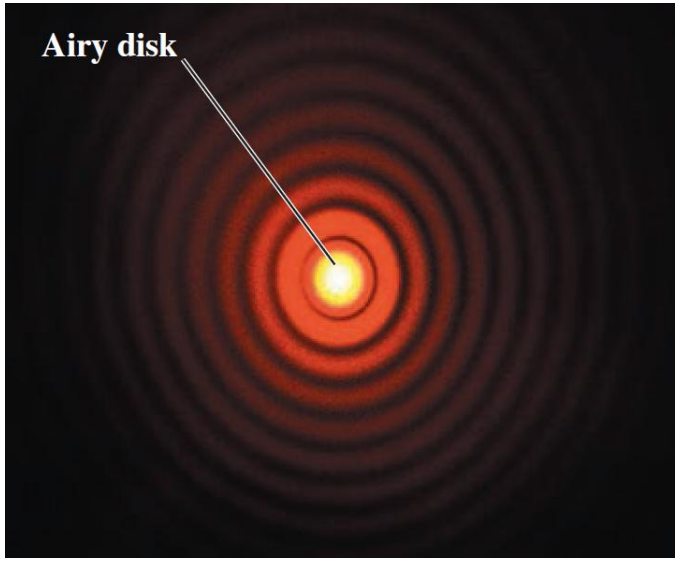
## Question 9

10 pts

X-rays of wavelength 0.2 nm incident at a grazing angle of 15.9° on an unknown crystal produces a first-order interference fringe. Calculate the interatomic spacing between the Bragg planes in nanometers.

# Diffraction by a Circular Aperture

- The central bright spot in the diffraction pattern of a circular aperture is called the Airy disk.
- We can describe the radius of the Airy disk by the angular radius $\theta_1$ of the first dark ring:




**Diffraction by a circular aperture:**

Angular radius of first dark ring = angular radius of Airy disk

$$\sin\theta_1 = 1.22\frac{\lambda}{D}$$

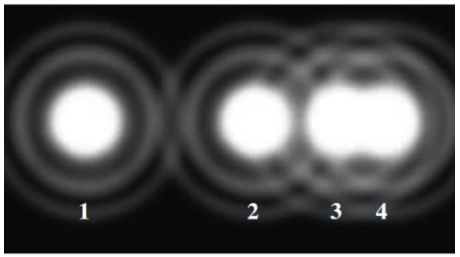
$\lambda$: Wavelength

$D$: Aperture diameter

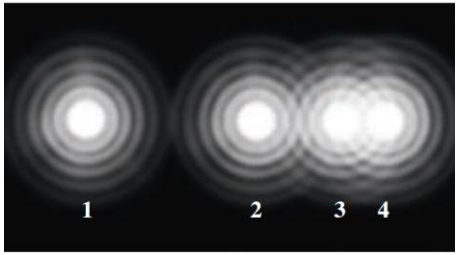
# Diffraction and Image Formation

- Diffraction limits the resolution of optical equipment, such as telescopes.
- The larger the aperture, the better the resolution.
- A widely used criterion for resolution of two point objects, is called **Rayleigh's criterion**:
  - Two objects are just barely resolved (that is, distinguishable) if the center of one diffraction pattern coincides with the first minimum of the other.

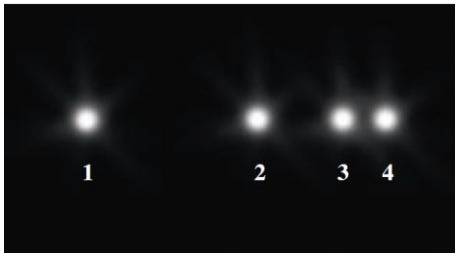
(a) Small aperture




(b) Medium aperture




(c) Large aperture



# Bigger Telescope, Better Resolution

- Because of diffraction, large-diameter telescopes, such as the VLA radio telescope below, give sharper images than small ones.

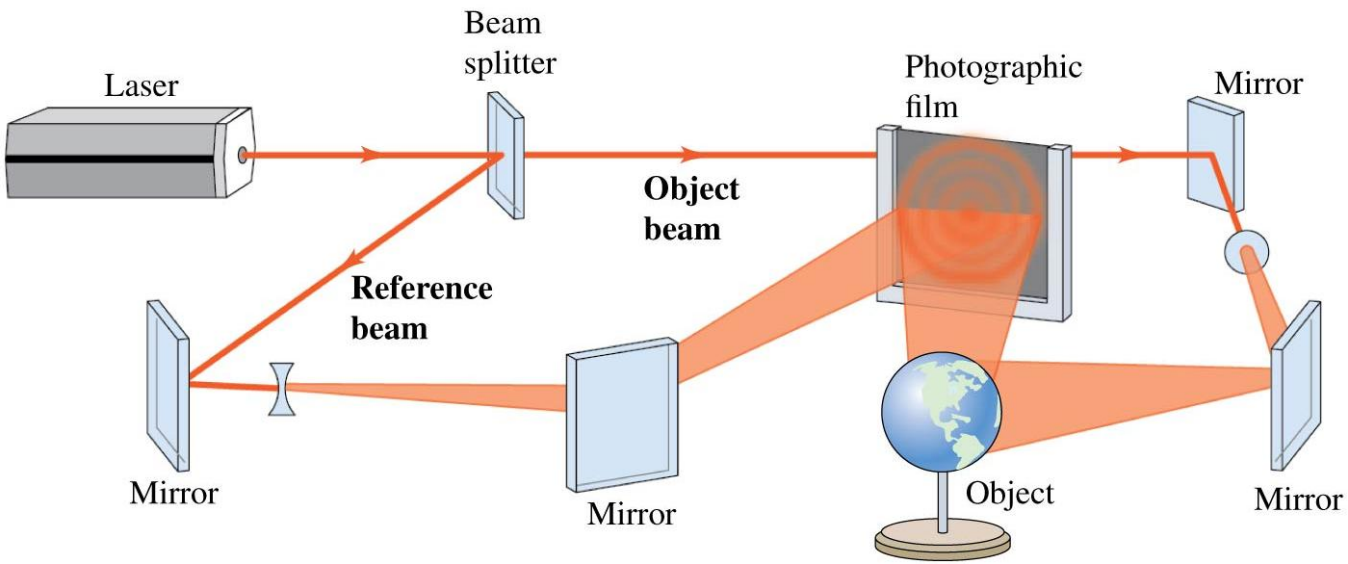
## Question 10

10 pts

Suppose a space telescope orbiting Mars has an aperture diameter of 5.6 cm. What minimum width crater (in meters) can it resolve in light of wavelength 530 nm if it is in an orbit 1,833 km from the planet's surface. Disregard atmospheric effects.

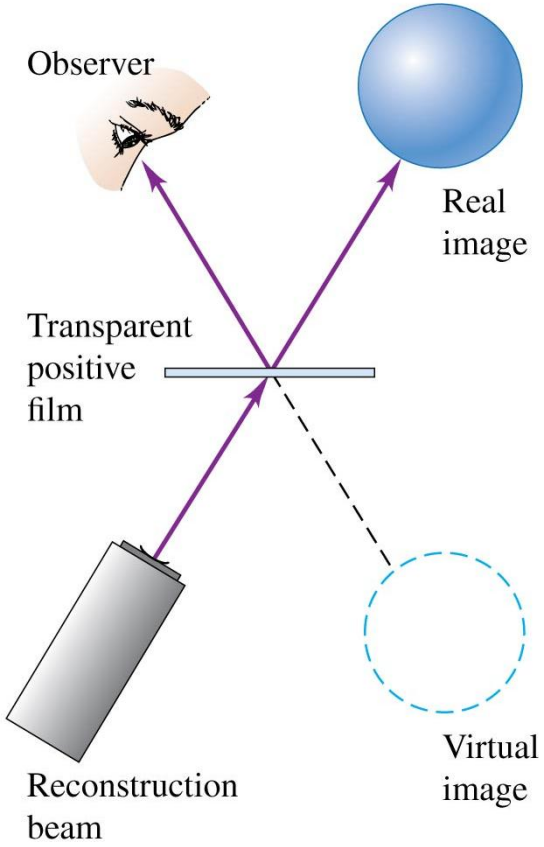
# What Is Holography?

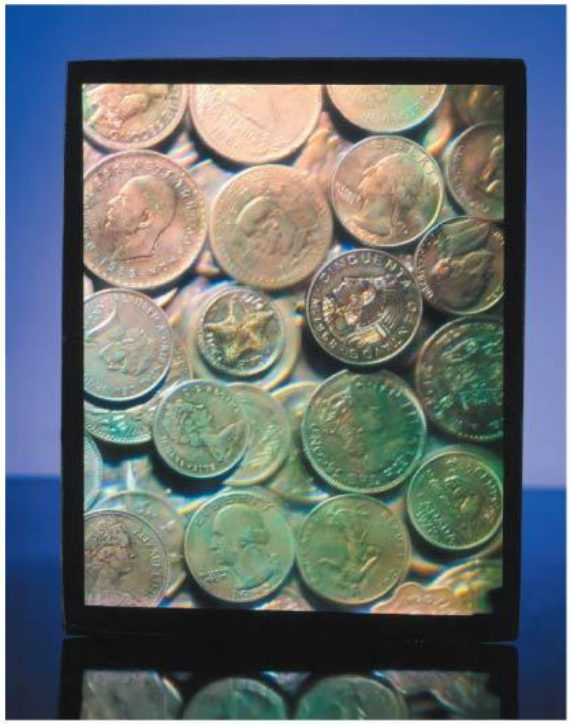
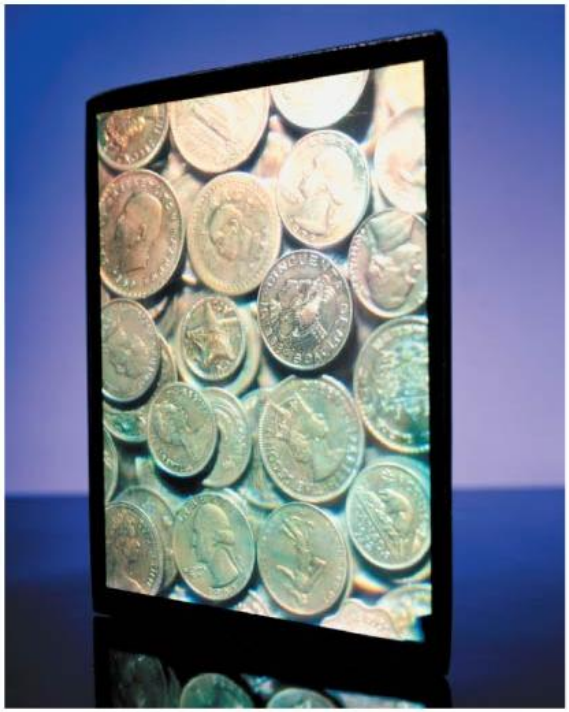
- By using a beam splitter and mirrors, coherent laser light illuminates an object from different perspectives.
- Interference effects provide the depth that makes a three-dimensional image from two-dimensional views.



# Viewing the Hologram

- A hologram is the record on film of the interference pattern formed with light from the coherent source and light scattered from the object.
- Images are formed when light is projected through the hologram.
- The observer sees the virtual image formed behind the hologram.



# An Example of Holography

- Shown below are photographs of a holographic image from two different angles, showing the changing perspective.